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United States Patent	12401321
Kind Code	B2
Date of Patent	August 26, 2025
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Bias circuit for crystal oscillator

Abstract

A bias circuit may generate a one or more bias outputs for a crystal oscillator. The bias circuit may include a first current mirror, a second current mirror and a third current mirror. A capacitor may be coupled between the second current mirror and the third current mirror. In operation, a first bias output may be generated by the third current mirror, and a second bias output may be generated by the second current mirror. The first bias output and second bias output may be coupled to a crystal oscillator.

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Appl. No.:	18/403193
Filed:	January 03, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250132728 A1	Apr. 24, 2025

Related U.S. Application Data

us-provisional-application US 63544795 20231019

Publication Classification

Int. Cl.: **H03B5/36 (20060101)**

U.S. Cl.:

CPC **H03B5/364** (20130101);

Field of Classification Search

CPC: H03B (5/364)

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Background/Summary

PRIORITY (1) This application claims priority to commonly owned U.S. Provisional Patent Application No. 63/544,795 filed Oct. 19, 2023, the entire contents of which are hereby incorporated by reference for all purposes.

FIELD OF THE INVENTION

(1) The present disclosure relates to a bias circuit for a crystal oscillator.

BACKGROUND

(2) Crystal oscillators may be driven with an input from an AC coupled driver circuit. Such examples may include high value resistors when using current levels below 100 nA. As one of various examples, resistor values may exceed 1 MΩ. Crystal oscillators with high value resistors may be large and expensive.

(3) Other examples may utilize a class-A or class-C amplifier which may represent an inefficient solution which may waste current.

(4) There is a need for a bias circuit which can operate at a low current bias point and eliminate the need for a high value resistor and provide a bias current for a crystal oscillator.

SUMMARY

(5) The examples herein enable a device for generating a bias current for a crystal oscillator.

(6) According to one aspect, a bias circuit includes a first current mirror comprising a current

source coupled to a first current mirror device. The current source may be coupled to a first node and a second node of the first current mirror device. The third node of the first current mirror device may be coupled to a first fixed voltage. A second current mirror may include a second current mirror device and a first cascode device. A first node and a second node of the first cascode device may be coupled together and comprise a current path. A first node of the second current mirror device may be coupled to a third node of the first cascode device. A second node of the second current mirror device may be coupled to the current source. A third node of the second current mirror device may be coupled to the first fixed voltage. A third current mirror may include a third current mirror device and a second cascode device. A first node of the second cascode device may be coupled to an output driver. A second node of the second cascode device may be coupled to the second node of the first cascode device. A third node of the second cascode device may be coupled to a first node of the third current mirror device. A second node of the third current mirror device may be coupled to the current source. A third node of the third current mirror device may be coupled to the first fixed voltage. The output driver includes a first node coupled to the first node of the second cascode device, a second node coupled to a first node of a capacitor, and a third node coupled to a second fixed voltage. The first node of the output driver may be a first bias output and the second node of the output driver may be a second bias output. The capacitor includes a first node coupled to a second node of the output driver and a second node coupled to the second node of the second cascode device.

(7) According to one aspect, a system includes a bias circuit to generate a first bias output and a second bias output based on a first current mirror. The first current mirror may include a current source and a first current mirror device. The system may include a crystal oscillator to receive input from the first bias output and the second bias output and to generate an oscillation output.

(8) According to one aspect, a method includes steps of: generating a reference current in a first current mirror, generating a first bias output, the first bias output based on the reference current, generating a second bias output, the second bias output based on the reference current, outputting the first bias output and the second bias output to a crystal oscillator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates one of various examples of a bias circuit.
- (2) FIG. 2 illustrates one of various examples of a crystal oscillator system.
- (3) FIG. 3 illustrates a timing diagram of a bias circuit and crystal oscillator system.
- (4) FIG. 4 illustrates a method of biasing a crystal oscillator.

DETAILED DESCRIPTION

- (5) FIG. 1 illustrates one of various examples of a bias circuit **100** for a crystal oscillator.
- (6) Bias circuit **100** may include current source **110**. Current source **110** may be a dependent current source or an independent current source. Current source **110** may be a current mirror or may be another type of current source not specifically mentioned. Current source **110** may generate a reference current. Current source **110** may be coupled to first current mirror device **111**. In one of various examples, current source **110** may be coupled to a first node and a second node of first current mirror device **111**. In one of various examples, first current mirror device **111** may comprise a metal-oxide semiconductor field-effect (MOSFET) device and current source **110** may be coupled to a drain node and a gate node of first current mirror device **111**. A third node of first current mirror device **111** may be coupled to first fixed voltage **180**. The third node of first current mirror device may comprise a source node. First fixed voltage **180** may be a positive supply voltage.
- (7) Current source **110** and first current mirror device **111** may comprise a first current mirror.
- (8) Bias circuit **100** may include second current mirror device **121** and first cascode device **122**. In

one of various examples, a first node of second current mirror device **121** may be coupled to first cascode device **122** and the first node of second current mirror device **121** may comprise a drain node of a MOSFET device. In one of various examples, a second node of second current mirror device **121** may be coupled to current source **110** and the second node of second current mirror device **121** may be a gate node of a MOSFET device. A third node of second current mirror device **121** may be coupled to first fixed voltage **180**, and the third node of second current mirror device **121** may comprise a source node of a MOSFET device. First fixed voltage **180** may be a positive supply voltage.

(9) Current path **120** may be coupled to first cascode device **122**. Current path **120** may be coupled to other circuitry not specifically mentioned or illustrated. In one of various examples, current path **120** may be coupled to a first node and a second node of first cascode device **122**. In one of various examples, first cascode device **122** may comprise a MOSFET device and current path **120** may be coupled to a drain node and a gate node of first cascode device **122**. A third node of first cascode device **122** may be coupled to the first node of second current mirror device **121**.

(10) Second current mirror device **121** and first cascode device may comprise a second current mirror.

(11) Bias circuit **100** may include third current mirror device **131**, second cascode device **132** and output driver **150**. A first node of output driver **150** may be coupled to a first node of second cascode device **132**. The first node of output driver **150** may comprise a drain node of a MOSFET device. The first node of output driver **150** may be first bias output **190**. In one of various examples, a second node of output driver **150** may be coupled to a first node of capacitor **140** and the second node of output driver **150** may be a gate node of a MOSFET device. The second node of output driver **150** may be second bias output **191**. In one of various examples, output driver **150** may be a MOSFET device and a third node of output driver **150** may be coupled to a second fixed voltage **199**. Second fixed voltage **199** may be a ground return. Second fixed voltage **199** may be a negative supply voltage. Second fixed voltage **199** may be a positive supply voltage.

(12) In one of various examples, the first node of output driver **150** may be coupled to a first node of second cascode device **132** and the first node of second current cascode device **132** may comprise a drain node of a MOSFET device. In one of various examples, a second node of second cascode device **132** may be coupled to the second node of first cascode device **122** and to a second node of capacitor **140**, and the second node of second cascode device **132** may be a gate node of a MOSFET device. A third node of second cascode device **132** may be coupled to a first node of third current source device **131**. The first node of third current source device **131** may comprise a drain node of a MOSFET device. A second node of third current mirror device **131** may be coupled to current source **110** and may comprise a gate node of a MOSFET device. A third node of third current mirror device may be coupled to first fixed voltage **180** and may comprise a source node of a MOSFET device.

(13) Third current mirror device **131** and second cascode device **132** may comprise a third current mirror.

(14) In operation, first current mirror device **111** and second current mirror device **121** may be sized such that first current source **110** may supply a current equal to twice the current flowing through second current mirror device **121**. In other examples, first current mirror device **111** and second current mirror device **121** may be sized such that first current source **110** may supply a current of a multiple of the current flowing through second current mirror device **121**. The lower current through second current mirror device **121** may bias second current mirror device **121** in the triode region.

(15) The maximum current flowing to first bias output **190** may be limited by the maximum current flowing through third current mirror device **131**. In one of various examples, third current mirror device **131** may be sized to be a larger size than second current mirror device **121** by a factor of N, and second cascode device **132** may be sized to be a larger size than first cascode device **122** by a

factor of N. In this example, the maximum current flowing to first bias output **190** may be $N \cdot I$, where I represents the value of current flowing through first current mirror device **111**. The factor N may be a real number value.

(16) In one of various examples, bias circuit **100** may create a high impedance at the second node of first cascode device **122**. In one of various examples, for a value of $I=5$ nA, an equivalent resistance of 100 M Ohms may be introduced at the second node of first cascode device **122**.

(17) In operation, first bias output **190** and second bias output **191** may be coupled to a crystal oscillator. The crystal oscillator may oscillate based on first bias output **190** and second bias output **191**.

(18) In the example illustrated in FIG. **1**, first current mirror device **111**, second current mirror device **121** and third current mirror device **131** are illustrated as p-channel metal-oxide semiconductor (PMOS) devices, but this is not intended to be limiting. In other examples, first current mirror device **111**, second current mirror device **121** and third current mirror device **131** may be n-channel metal-oxide semiconductor (NMOS) devices, bipolar junction transistors (BJTs) or other transistor types not specifically mentioned.

(19) In the example illustrated in FIG. **1**, first cascode device **122**, second cascode device **132** are illustrated as PMOS devices, but this is not intended to be limiting. In other examples, first cascode device **122** and second cascode device **132** may be NMOS devices, bipolar junction transistors (BJTs) or other transistor types not specifically mentioned.

(20) In the example illustrated in FIG. **1**, output driver **150** is illustrated as an NMOS device, but this is not intended to be limiting. In other examples, output driver **150** may be a PMOS device, a BJT, or other transistor type not specifically mentioned. In examples with a PMOS output driver **150**, first cascode device **122** and second cascode device **132** may be NMOS devices, and first current mirror device **111**, second current mirror device **121** and third current mirror device **131** may be NMOS devices.

(21) FIG. **2** illustrates one of various examples of a crystal oscillator system **200**.

(22) Bias circuit **220** may generate a first bias output **290**. First bias output **290** may represent one of various examples of first bias output **190** as described and illustrated in reference to FIG. **1**.

(23) Bias circuit **220** may generate a second bias output **291**. Second bias output **291** may represent one of various examples of second bias output **191** as described and illustrated in reference to FIG. **1**.

(24) First bias output **290** may be coupled to crystal oscillator **250**. Second bias output **291** may be coupled to crystal oscillator **250**. Crystal oscillator **250** may generate an oscillator output at first bias output **290** based at least on first bias output **290** and second bias output **291**.

(25) FIG. **3** illustrates one of various examples of a timing diagram of a crystal oscillator system.

(26) Trace **310** may represent a bias current. Trace **310** may represent one of various examples of a current supplied by current source **110** as described and illustrated in reference to FIG. **1**. Prior to time **390**, the bias current may be zero. At time **390**, a bias current may be generated, indicated by the change in trace **310**. The change in trace **310** is intended to indicate a bias current at a non-zero level. The height of trace **310** should not be interpreted to represent a specific numerical value of voltage.

(27) Trace **320** may represent a first bias output. Trace **320** may represent one of various examples of first bias output **190** as described and illustrated in reference to FIG. **1**. Prior to time **391**, the first bias output may be at a zero voltage level. At time **391**, the first bias output may transition to a non-zero value based on the bias current, indicated by the change in trace **320**. The change in trace **320** is intended to indicate the first bias output at a non-zero level. Trace **320** may be coupled to a crystal oscillator, and the crystal oscillator may begin to oscillate based on the level of trace **320**. The oscillation may be a sinusoidal signal.

(28) Trace **330** may represent a second bias output. Trace **330** may represent one of various examples of second bias output **191** as described and illustrated in reference to FIG. **1**. Prior to time

391, the second bias output may be at a zero voltage. At time **391**, the second bias output may transition to a non-zero value based on the bias current, indicated by the change in trace **330**. The change in trace **330** is intended to indicate the second bias output at a non-zero level. Trace **320** may be coupled to a crystal oscillator.

(29) The relative timing between time **390**, time **391** and time **392** is not intended to be limiting. The time between time **390** and time **391** may be larger or smaller than illustrated in FIG. 3. The time between time **391** and time **392** may be larger or smaller than illustrated in FIG. 3. In one of various examples, trace **320** and trace **330** may transition to non-zero values at the same time or trace **330** may transition to a non-zero value after trace **320** transitions to a non-zero value.

(30) FIG. 4 illustrates a method to bias a crystal oscillator circuit.

(31) At operation **410**, a reference current may be generated.

(32) At operation **420**, a first bias output may be generated, the first bias output at least based on the reference current. The first bias output may be based on a current one-half the value of the reference current.

(33) At operation **430**, a second bias output may be generated, the second bias output at least based on the reference current.

(34) At operation **440**, the first bias output and second bias output may be coupled to a crystal oscillator to drive the crystal oscillator.

Claims

1. A bias circuit comprising: a first current mirror comprising a current source coupled to a first current mirror device, the current source coupled to a first node and a second node of the first current mirror device, wherein the third node of the first current mirror device is coupled to a first fixed voltage; a second current mirror comprising a second current mirror device and a first cascode device, wherein: a first node and a second node of the first cascode device are coupled together and comprise a current path; a first node of the second current mirror device is coupled to a third node of the first cascode device; a second node of the second current mirror device is coupled to the current source; a third node of the second current mirror device is coupled to the first fixed voltage; a third current mirror comprising a third current mirror device and a second cascode device, wherein: a first node of the second cascode device is coupled to an output driver; a second node of the second cascode device is coupled to the second node of the first cascode device; a third node of the second cascode device is coupled to a first node of the third current mirror device; a second node of the third current mirror device is coupled to the current source; a third node of the third current mirror device is coupled to the first fixed voltage; wherein the output driver comprises a first node coupled to the first node of the second cascode device, a second node coupled to a first node of a capacitor, and a third node coupled to a second fixed voltage, the first node of the output driver comprising a first bias output and the second node of the output driver comprising a second bias output; and wherein the capacitor comprises a first node coupled to a second node of the output driver and a second node coupled to the second node of the second cascode device.
2. The bias circuit as claimed in claim 1, the first fixed voltage comprising a positive supply voltage.
3. The bias circuit as claimed in claim 1, the second fixed voltage comprising a negative supply voltage.
4. The bias circuit as claimed in claim 1, the second fixed voltage comprising a ground return.
5. The bias circuit as claimed in claim 1, the second fixed voltage comprising a positive supply voltage.
6. The bias circuit as claimed in claim 1, the current through the current source equal to twice the current through the second current mirror device and the first cascode device.
7. The bias circuit as claimed in claim 1, the size of the third current mirror device a multiple of the

size of the second current mirror device.

8. The bias circuit as claimed in claim 1, the size of the second cascode device a multiple of the size of the first cascode device.

9. A crystal oscillator system comprising: a bias circuit to generate a first bias output and a second bias output based on a first current mirror, the first current mirror comprising a current source and a first current mirror device; a crystal oscillator to receive input from the first bias output and the second bias output and to generate an oscillation output; the bias circuit comprising a second current mirror to receive input from the current source and a third current mirror to receive input from the current source; the third current mirror comprising a third current mirror device coupled to a second cascode device and an output driver coupled to the second cascode device, and a capacitor coupled between a gate of the output driver and the second cascode device.

10. The crystal oscillator system as claimed in claim 9, the second current mirror comprising a second current mirror device coupled to a first cascode device.

11. The crystal oscillator system as claimed in claim 9, the first bias output at a zero voltage level at a first time and the first bias output at a non-zero voltage level at a second time, the second time after the first time.

12. The crystal oscillator system as claimed in claim 9, the second bias output at a zero voltage level at a first time and the second bias output at a non-zero voltage level at a second time, the second time after the first time.

13. The crystal oscillator system as claimed in claim 9, at a predetermined time, the first bias output at a non-zero voltage level and the second bias output at a zero voltage level.

14. The crystal oscillator system as claimed in claim 9, at a predetermined time, the first bias output at a zero voltage level and the second bias output at a non-zero voltage level.

15. The crystal oscillator system as claimed in claim 9, the first bias output and the second bias output to transition to a non-zero voltage level simultaneously.

16. The crystal oscillator system as claimed in claim 9, the crystal oscillator to oscillate based on the first bias output.

17. The crystal oscillator system as claimed in claim 9, the crystal oscillator to oscillate based on the second bias output.

18. A method comprising: generating a reference current in a first current mirror; generating a first bias output, the first bias output based on the reference current, the first bias output based on a current in a cascode device coupled to a current mirror device directly coupled to the reference current; generating a second bias output, the second bias output based on the reference current; and outputting the first bias output and the second bias output to a crystal oscillator, further comprising a capacitor coupled between the second bias output and the cascode device.
